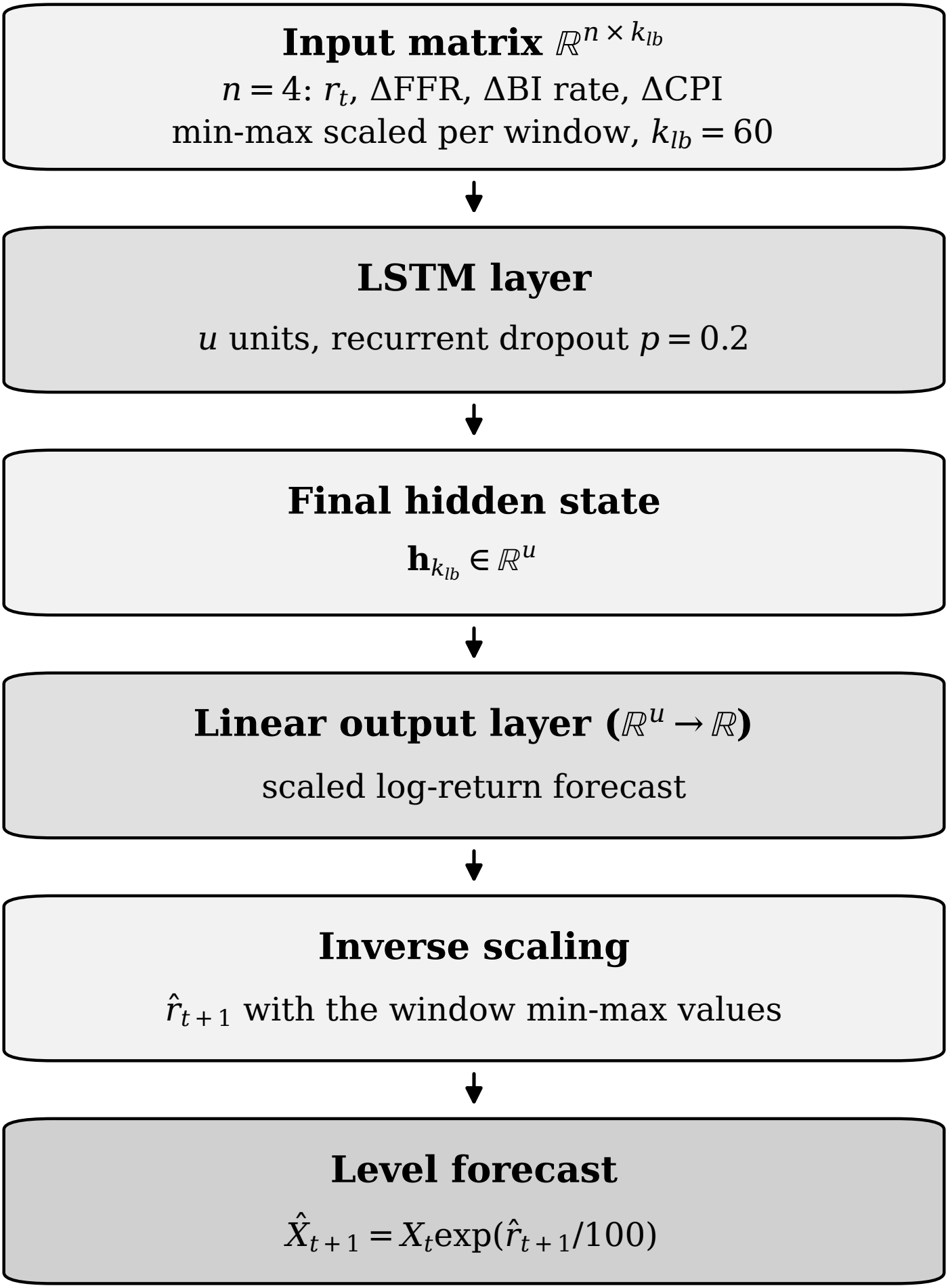
Input matrix $\mathbb{R}^{n \times k_{lb}}$
$n = 4$: r_t, ΔFFR, ΔBI rate, ΔCPI
min-max scaled per window, $k_{lb} = 60$
LSTM layer
u units, recurrent dropout $p = 0.2$
Final hidden state
$\mathbf{h}_{k_{lb}} \in \mathbb{R}^u$
Linear output layer ($\mathbb{R}^u \to \mathbb{R}$)
scaled log-return forecast
Inverse scaling
$\hat{r}_{t+1}$ with the window min-max values
Level forecast
$\hat{X}_{t+1} = X_t \exp(\hat{r}_{t+1}/100)$